EMG4_analysis_behavioral_data.pdf

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Contents

R packages

Embedded Ratings

```
# number of NA responses
sum(is.na(embedded_data$embed_rating))
```

```
## [1] 39
```

```
table(embedded_data$embed_rating)
```

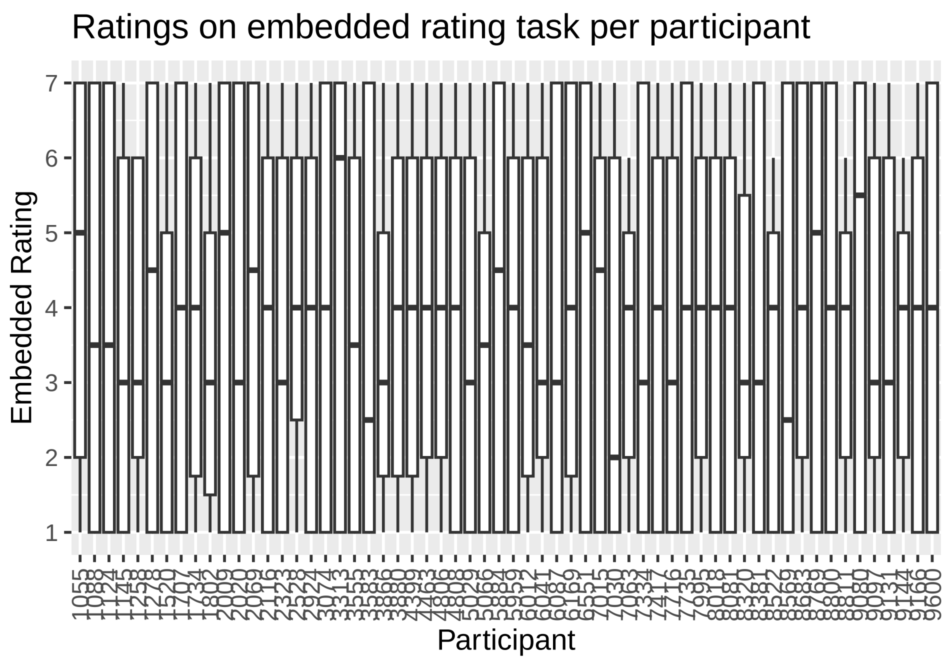
```
##
##    1    2    3    4    5    6    7
## 1158  450  253  165  385  601  789
```

RT ratings
nb:remove NA values from trials with missing ratings

```
embedded_ratings <- embedded_data %>%
  select(sjCode, Markertrial, embed_rating, embed_RT) %>%
  filter(!is.na(embed_rating))
head(embedded_ratings)
```

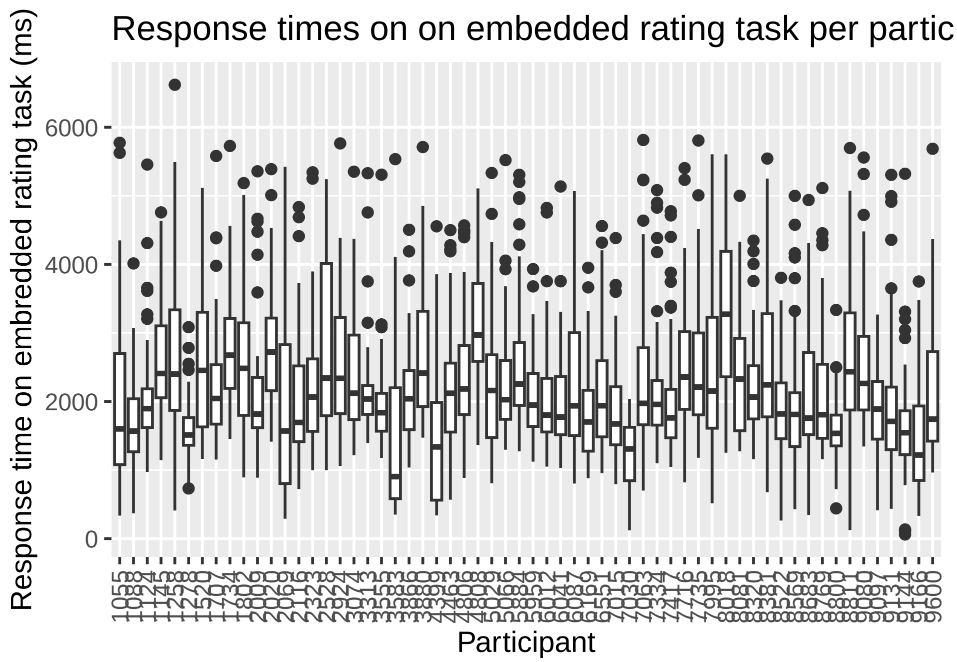
```
## # A tibble: 6 x 4
##   sjCode Markertrial embed_rating embed_RT
##    <dbl>       <dbl>        <dbl>    <dbl>
## 1   1055          52            7    1754.
## 2   1055          49            7    1738.
## 3   1055          55            7    1563.
## 4   1055           8            1    1230.
## 5   1055          10            1    2231.
## 6   1055          57            7     900.
```

ratings per participant



Rating times

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    60.7  1541.5  1972.7  2200.8  2680.9  6619.3
```



Fairness ratings

```r
fairness_data <- read_csv(paste0(getwd(), "/Qualtrics/fairness_all_EMG4_final.csv")) %>%
  rename(fairness_rating = F_rating)
```

```
## Rows: 3685 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (3): sjCode, Trial, F_rating
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# number of NA responses
sum(is.na(fairness_data$fairness_rating))
```

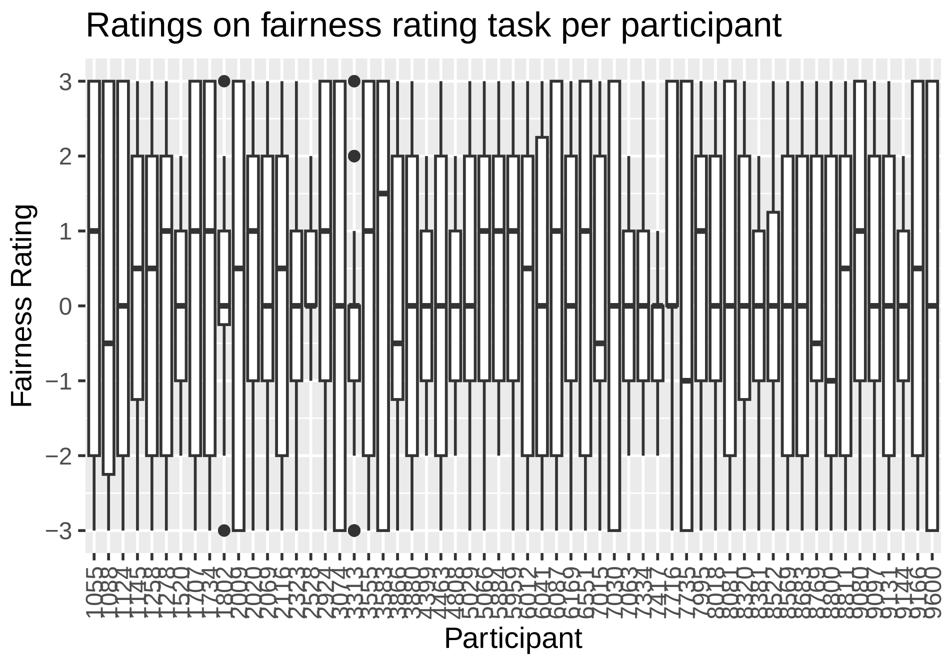
```
## [1] 0
```

```r
table(fairness_data$fairness_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
## 408 525 599 442 434 583 694
```

```r
fairness_ratings <- fairness_data %>%
  filter(!is.na(fairness_rating))
table(fairness_ratings$fairness_rating)
```

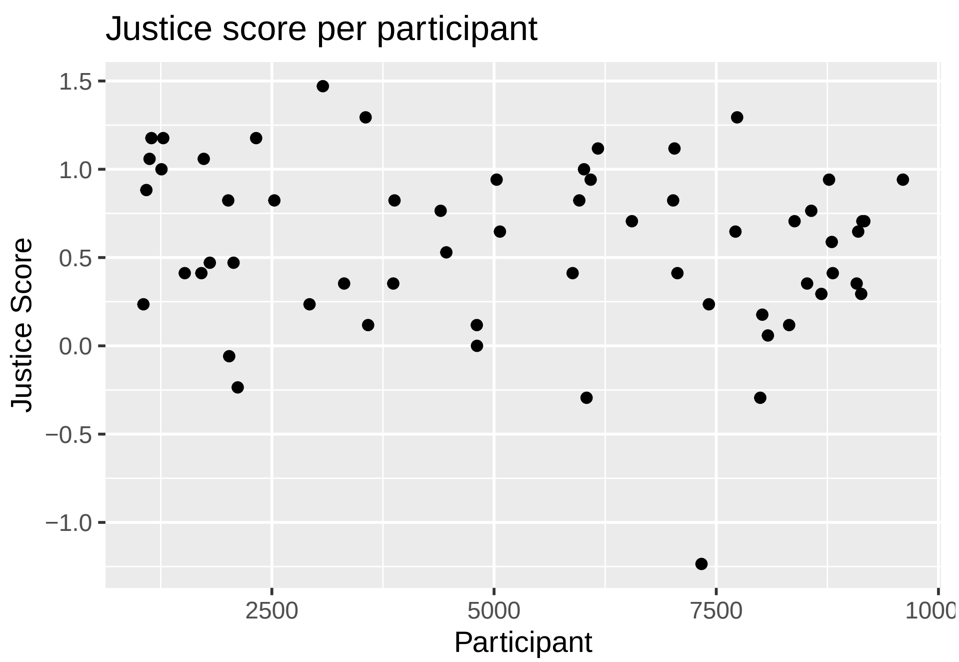
```
##
##   -3  -2  -1   0   1   2   3
## 408 525 599 442 434 583 694
```



Justice scores

```r
# number of NA responses
summary(justice_means$justice_score)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1.2353  0.2941  0.6471  0.5716  0.9412  1.4706
```



comprehension questions check

check number of trials with incorrect comprehension responses per participant

```
# check how many questions were correct (1) and incorrect (1)
table(comprehension_data$comp_correct)
```

```
##
##   0   1
## 150 559
```

```
#  check comprehension questions, for keys used (1 and 7 were the response keys)
table(na.omit(comprehension_data$comp_response))
```

```
##
##   1   2   7
## 352   7 350
```

```
# check number of missed responses
NA_questions <- comprehension_data %>%
  select(sjCode, questionnr, comp_response) %>%
  filter(is.na(comp_response)) %>%
  group_by(sjCode, questionnr) %>%
  summarise(Freq = n())
```

```
## `summarise()` has grouped output by 'sjCode'. You can override using the
## `.groups` argument.
```

```
head(NA_questions) # only 6 missings
```

```
## # A tibble: 6 x 3
## # Groups:   sjCode [5]
##   sjCode questionnr  Freq
##    <dbl>      <dbl> <int>
## 1   1055          4     1
## 2   2009         12     1
## 3   2528          7     1
## 4   2528          8     1
## 5   3583          5     1
## 6   4399          1     1
```

```
# check number of erroneous key 2 used
Resp2_questions <- comprehension_data %>%
  select(sjCode, questionnr, comp_response) %>%
  filter(comp_response==2) %>%
  group_by(sjCode, questionnr) %>%
  summarise(Freq = n())
```

```
## `summarise()` has grouped output by 'sjCode'. You can override using the
## `.groups` argument.
```

```
head(Resp2_questions) # only 6 erroneous keys used
```

```
## # A tibble: 6 x 3
## # Groups:   sjCode [5]
##   sjCode questionnr  Freq
##    <dbl>      <dbl> <int>
## 1   1145          3     1
## 2   1145          8     1
## 3   2069          8     1
## 4   3880          3     1
## 5   4463          8     1
## 6   4808          3     1
```

```
# On 6 occasions a response 2 has been given,
# this is not a valid response and needs to be corrected to NA.
comprehension_data_clean <- comprehension_data %>%
  select((sjCode:comp_correct)) %>%
  mutate(comp_RT = if_else(comp_response==2, NA, comp_RT))%>%
  mutate(comp_correct = if_else(comp_response==2, NA, comp_correct))
```

number and percentage of comprehension questions answered correctly per participant

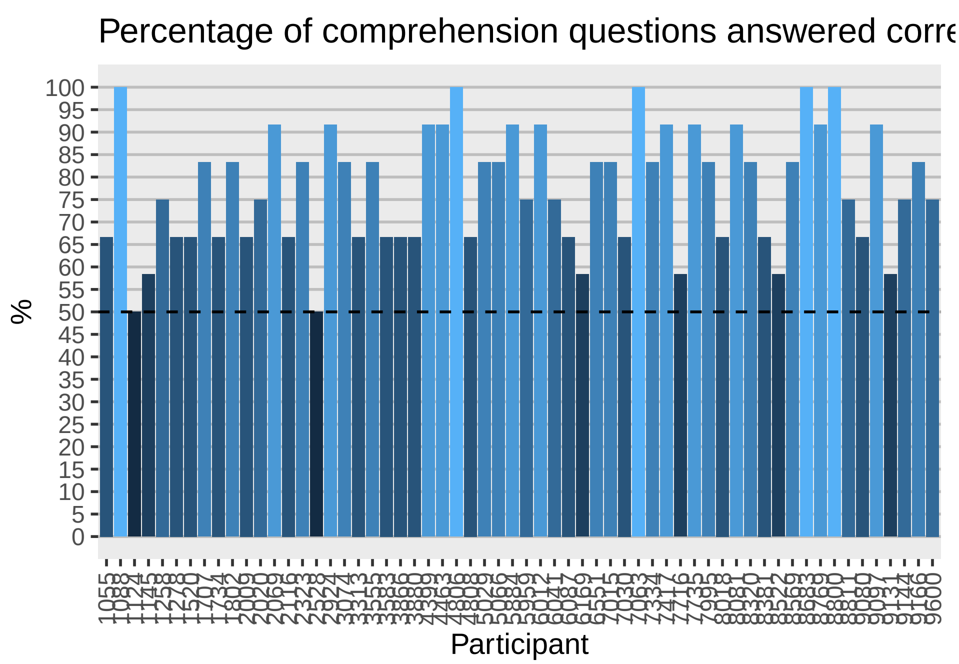
```
corr_comp_perc <- comprehension_data_clean %>%
  select(sjCode, comp_correct) %>%
  group_by(sjCode) %>%
  summarise(sum_CorrComp = sum(comp_correct, na.rm = TRUE)) %>%
  mutate(comp_percentage = sum_CorrComp/12*100)

head(corr_comp_perc)
```

```
## # A tibble: 6 x 3
##   sjCode sum_CorrComp comp_percentage
##    <dbl>        <dbl>           <dbl>
## 1   1055            8            66.7
## 2   1088           12           100
## 3   1124            6            50
## 4   1145            7            58.3
## 5   1258            9            75
## 6   1278            8            66.7
```

```
corr_comp_perc %>%
  filter(comp_percentage < 60.5)
```

```
## # A tibble: 7 x 3
##   sjCode sum_CorrComp comp_percentage
##    <dbl>        <dbl>           <dbl>
## 1   1124            6            50
## 2   1145            7            58.3
## 3   2528            6            50
## 4   6169            7            58.3
## 5   7716            7            58.3
## 6   8522            7            58.3
## 7   9131            7            58.3
```

create barplot:



check RT times on comprehension questions

```r
summary(comprehension_data_clean$comp_RT)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    1690    2595    3053    3239    3639    6615      18
```

```r
RT_comp <- comprehension_data_clean %>%
  drop_na() %>%
  arrange(comp_RT)

head(RT_comp)
```

```
## # A tibble: 6 x 6
##   sjCode listnr questionnr comp_response comp_RT comp_correct
##    <dbl>  <dbl>      <dbl>         <dbl>   <dbl>        <dbl>
## 1   9144      2          2             1   1690.            1
## 2   3313      2          5             7   1690.            1
## 3   9144      2          6             1   1721.            1
## 4   1088      3         11             1   1728.            1
## 5   5029      2          6             1   1745.            1
## 6   9144      2          7             1   1777.            1
```

```r
barplot_RT_comp <- ggplot(RT_comp, aes(x = as.character(sjCode), y = comp_RT)) +
  geom_boxplot() +
  scale_y_continuous(n.breaks = 10)  +
  labs(title = "Response time of comprehension question answers per participant",
       x = "Participant", y = "Response time (ms)") +
  theme(axis.text.x= element_text(angle = 90, vjust = 0.5))

barplot_RT_comp
```